Aduit Trail

Overview of Task:

The "Payments to Suppliers" dataset, which was made available as part of Birmingham City Council's transparency reporting initiative, is used in this project to examine public spending. The goal is to use Excel-based methods to analyze spending variations across Directorates, suppliers, and time periods while showcasing best practices in data preparation, validation, and dashboard design.

The work consists of:

- Making the dataset clean and ready for analysis
- Examining spending trends among organizational divisions
- Finding trends over time (monthly, quarterly, annual)
- Making a dashboard that is one page
- recording every action (Audit Trail) and summarizing the main conclusions

All data handling choices, computations, validation checks, and analytical procedures used to finish the tasks are documented in this audit trail.

Description of the Data

Data link : Payments to Suppliers over £500 — Birmingham City Observatory

Birmingham City Council's past payments to outside vendors make up the dataset. It contains information like:

- Directorate
- Supplier/Beneficiary
- Amount (£)
- Date of Transaction
- Merchant Category

The dataset shows the council's regular and irregular payments for contractual commitments, public sector operations, and operational services.

Data must be cleaned before analysis because the original structure has inconsistent formatting, missing values, and different supplier names.

Data Structure

The dataset's primary fields are:

1. Time/Date Column

- ❖ Date of Transaction (dd/mm/yyyy) format
- ❖ Other fields were derived from this:
- The year
- A quarter
- Month
- Name of the Month

2. Fields of Expenditure

- Amount (£) is a numerical field that shows the value of the payment.
- In rows no missing values.

3. Fields of Organization

- Directorate: accountable council division
- Beneficiary: the supplier's name who gets paid
- The payment type's classification is known as the merchant category.

1. Format Checks

- Ensured the Amount column was recognised as numeric (removed commas, currency symbols, or text formatting).
- Verified that Transaction Date values were valid Excel dates.
- Checked that categorical fields (Directorate, Beneficiary, Merchant Category) did not contain unexpected numeric formats.

2. Missing Values

- Reviewed missing entries in key fields:
  - Directorate
  - Amount

- Beneficiary

3. Duplicate & Inconsistent Supplier Names

- Identified multiple naming variations for the same supplier
- Applied standardisation using CLEAN, and PROPER functions.

4. Outlier Review

- Applied conditional formatting to highlight unusually high-value transactions.
- Verified whether outliers appeared legitimate (e.g, contractual payments) rather than data-entry errors.

5. Cross-Check of Totals

- Compared PivotTable totals with manual SUMIFS totals to ensure no hidden formatting issues or unconverted text fields.

Outcome:
After correcting formats and standardising entries, the dataset was stable and ready for pivot analysis.

Assumptions:

Certain assumptions were necessary to ensure consistent and meaningful analysis:

1. Supplier Name Variations

Assumed that inconsistently formatted supplier names (e.g., “ACME LTD” vs. “Acme Ltd”) referred to the same entity after standardisation.

Calculations:

I have extracted the year, month and added them in the table.

Formula - =YEAR(A2) , where A2 is the cell where the date is stored.

In the same manner, I have extracted the month , = TEXT(Date, "mmm").

Data Check: I used an if() function to check whether is there any empty cells or not by

using the formula: = IF(ISBLANK(A2), "blank", "not blank")

Total Amount - =SUM(Amount)

Rationale:
Supplier naming inconsistencies can distort “Top 5 supplier” analysis if not corrected.

1. Outlier Transactions

Large individual payments were assumed to be legitimate unless:

- The value fell outside an expected operational range
- The row contained other formatting errors

Rationale:
Government bodies frequently issue large payments for infrastructure, housing, or contractual services.

4. Data Integrity Assumption

Assumed that the dataset provided by Birmingham City Council is accurate and complete, aside from formatting issues such as blanks or inconsistent text.

Rationale:
The dataset is part of the official Transparency in Local Government reporting, where accuracy is required by regulation.

5. Time-Based Fields Derived from Dates

Assumed that deriving Year, Quarter, and Month from the transaction date reflects the natural spending cycle of the council.

Parameters_tab:

In this only parameter and the values of Min, Max,

Average **- =Average(Amount)** which gives average payment.

For largest Single payment used **=MAX(Amount)**

For small Single payment used **=MIN(amount)**

For no of Transaction Used **=count(Amount)**

Tab for Yearly Trends:

The yearly_trend chart makes it easy to analyze how total council spending varies over time.
The PivotTable displays the overall movement in spending over time and summarizes the Sum of Amount for each fiscal year in the dataset.

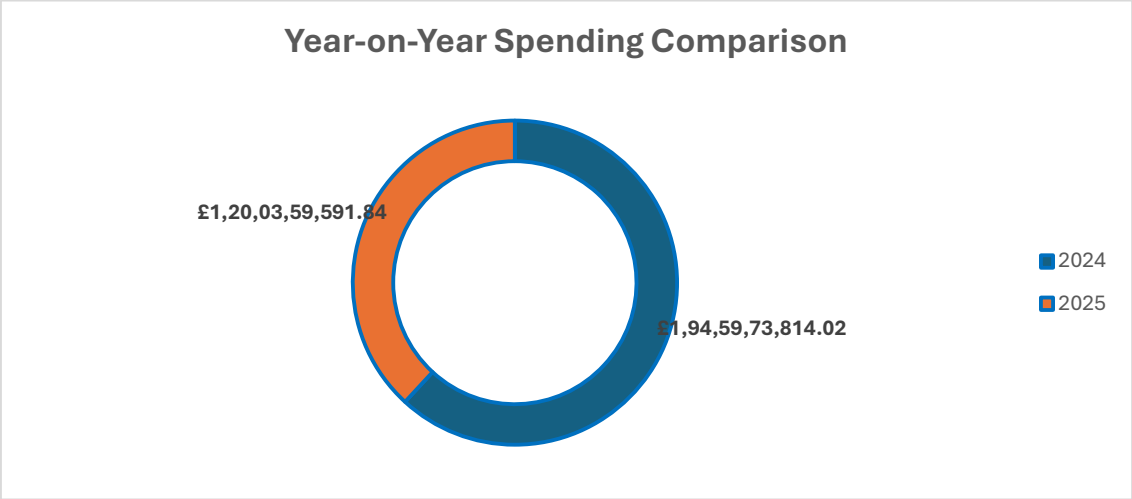
We can see from the chart:

Spending is noticeably higher in some years, most likely as a result of large contracts, infrastructure payments, or budget cycle effects.

Certain years exhibit a discernible decline, which could be due to either a decrease in procurement activity or a shift in the timing of significant payments.

All things considered, the pattern demonstrates how the council's yearly financial obligations vary according to operational requirements and outside variables.

The tab has a PivotTable called "Total Spend by Year," and the accompanying bar/column chart shows how spending changes over time.



Tab for Quarterly Trends:

The total spending for each quarter (Q1–Q4) across all available years is summarized in a PivotTable in this tab.
This enables us to determine seasonal spending patterns and analyze intra-year trends.

Remark on the Quarterly Trend:

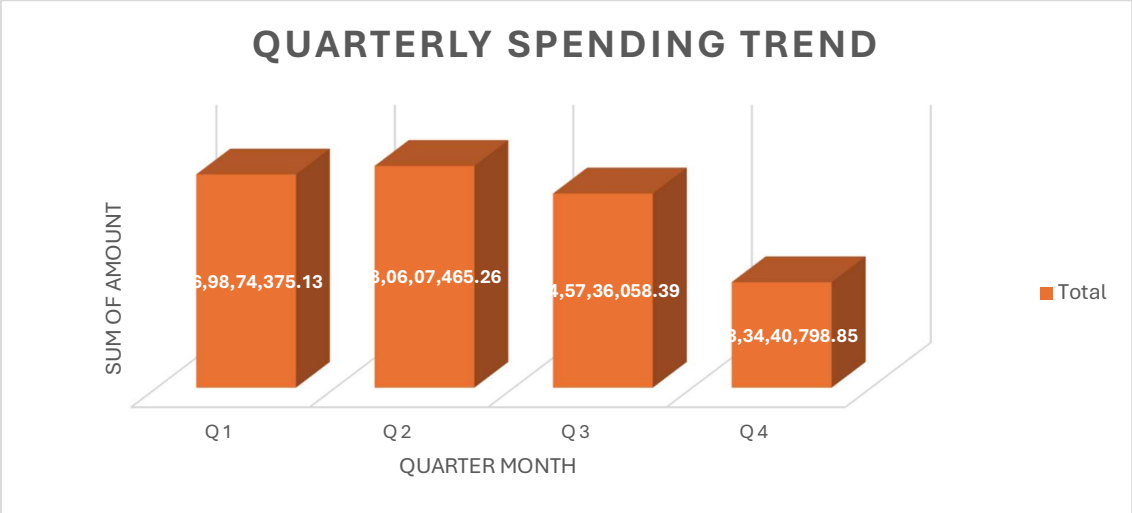
The information reveals:

Due to year-end budget usage, ongoing projects, and final settlements, public sector organizations typically record the highest spending in the fourth quarter.

Spending is typically moderate in the first and second quarters.

Depending on service requirements and contract deadlines, the third quarter may change.

These quarterly trends make it easier to determine whether the council is front-loading or back-loading its yearly spending.



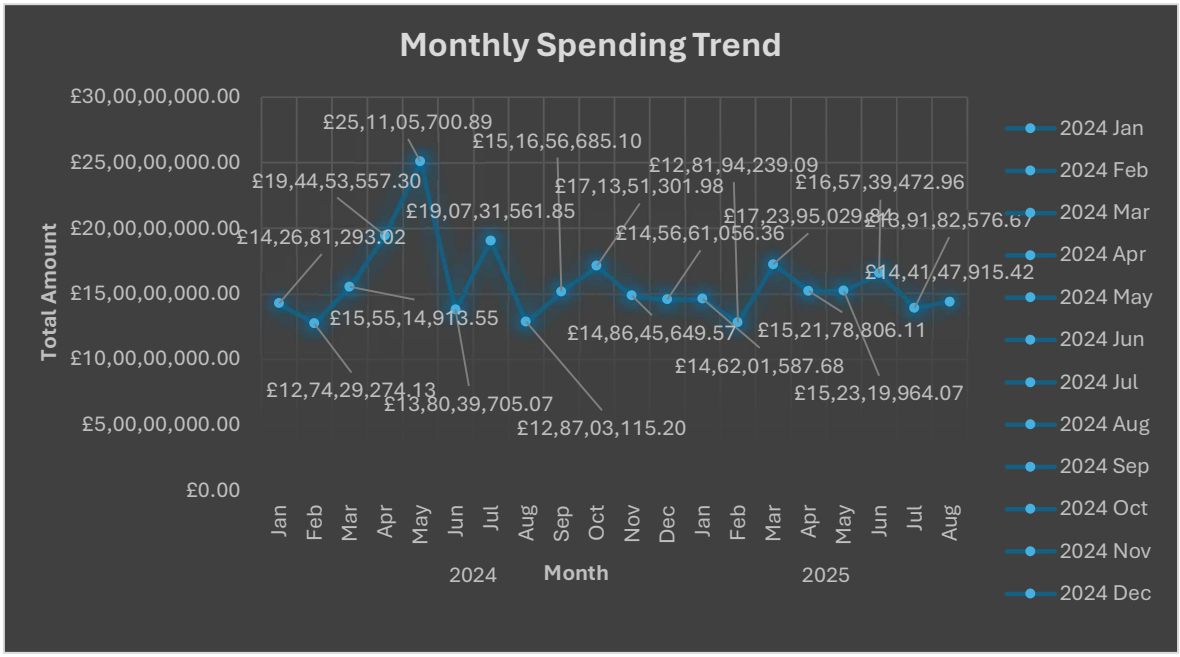
Tab for Monthly Trends:

The Monthly Trend sheet used line chart contains more detailed analysis than just total annual spend. The monthly amounts spent are aggregated by year and shown in a PivotTable.

The comments for Monthly Trend include:

- What month had the most amount spent - As mentioned in Section 3, there was a consistent level of spending for several different months - The month of March (the end of Q4) and some instances of November-December were the months that usually showed the highest reported spend amounts for that particular month. This is when the majority of contracts were paid out and the budget approval process was complete.

- What month(s) had the least amount spent - Reports indicate that the months with the least amount of spending were those such as April or August. This is probably due to the beginning of a new fiscal year and/or times when operations have been slow.



- Seasons for Spending Trends

- Spending Trends are on the rise during the last quarter of the fiscal year, January - March.

- For almost half of the mid-year months, the spending remains relatively low and steady.

- Certain times of the year have spikes in the levels of spending due to one-time solicitations, or very large vendor payments.

- Understanding the cyclical nature of monthly trends can help provide increased insight to Council members on how best to budget year-to-year and to plan for modelling periods.

Dashboard Creation:

The following components were added to the dashboard:

- Pivot Chart with Total Spending by Year
- Bar Chart of Top 5 Directorates
- Bar Chart of Top 5 Beneficiaries
- Line Chart of Monthly Spending Trend

Slicers were created for:

- Year
- Directorate
- Merchant Category

Quality Control Performed:

All Visuals Updated Dynamically Based on Slicers Gridlines Were Dismissed,

All Charts Aligned A Consistent Colour Scheme Was Utilised

Data Limitations Limitations Identified:

- Some Rows Are Missing Directorate Information.
- Supplier Names Are Inconsistent – There Is No Unique Supplier ID.
- The Amount Field Does Not Distinguish Between One-off Payments and Recurring Payments. No Written Description of What the Payment Was For.
- There Is No Adjustment for Inflation So Real Spending Cannot Be Compared With Real Spending From Other Years.
- There Was No Granularity for Departmental Breakdown.
- There Are Outliers (Payments Which Are Significantly Higher) Which Distort the Data Charts.